

PE-EC701B — Satellite Communication

Semester VII • 3L:0T:0P • 3 Credits

Module 3 Exam Notes: Satellite Sub-systems

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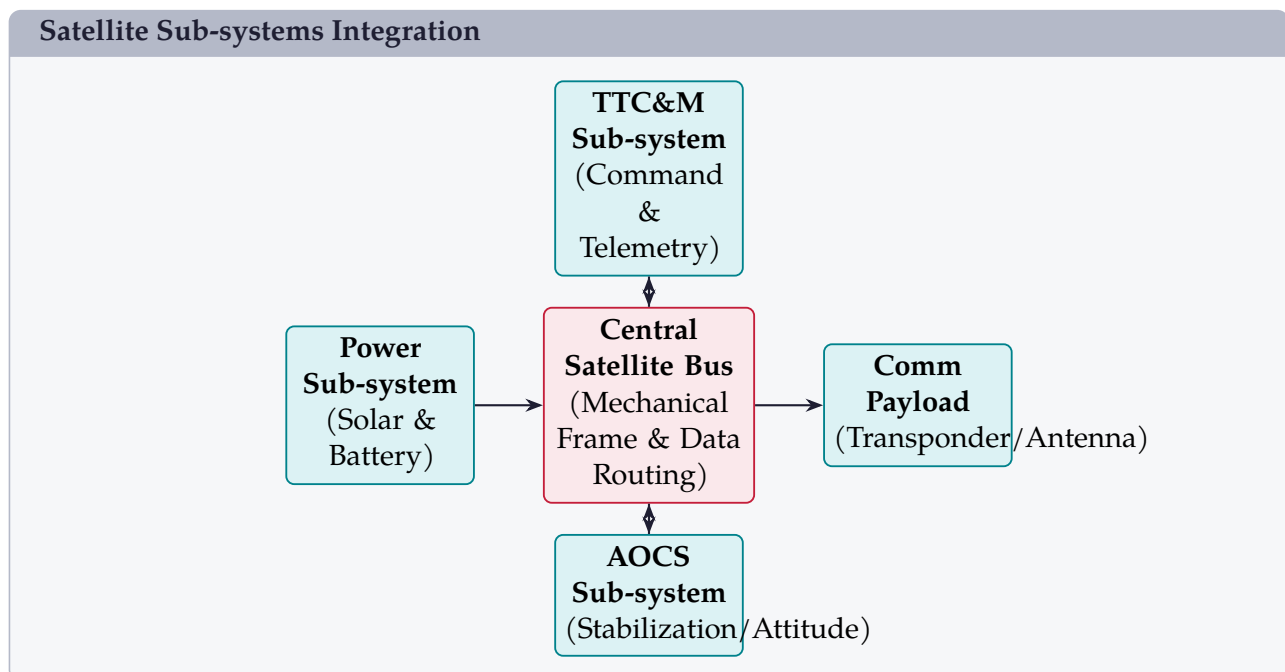
1. Satellite Sub-systems Architecture

A spacecraft consists of two primary operational components: the **Payload** (which performs the primary communication task) and the **Bus** (which provides the physical support infrastructure, power, stabilization, and control necessary for the payload to function).

Satellite Bus & Payload: The **payload** refers to the communication transponders and antennas. The **bus** refers to the physical support systems including the structural frame, thermal shielding, power generation, propulsion, stabilization (AOCS), and monitoring (TTC&M).

1.1 Sub-systems Overview

The subsystems must work in synchronization to maintain the satellite's health and functionality. The relationship between the systems is mapped below:



2. Telemetry, Tracking, Command, and Monitoring (TTC&M)

The TTC&M sub-system establishes a constant bi-directional radio control link between the satellite and the ground control station (Earth Station).

- **Telemetry (Transmission from Space to Earth):** Gathers diagnostic sensor data onboard the satellite (such as battery voltage, solar cell current, internal temperature, and gas pressure) and transmits this status information to Earth.
- **Tracking (Determination of Position):** Measures slant range, angular coordinates, and Doppler shift of the satellite's beacon to compute its precise orbital position, ensuring it has not drifted.

- **Command (Reception from Earth to Space):** Receives control instructions from the ground station to execute tasks like firing thrusters for orbital correction, switching transponder channels, or deploying solar arrays.
- **Monitoring:** Continuous checking of payload performance and checking for anomalies.

3. Attitude and Orbit Control System (AOCS)

The AOCS is responsible for keeping the satellite in its correct orbital slot (station-keeping) and ensuring its antennas and solar panels are pointing in the correct directions (attitude control).

- **Attitude Control:** Stabilizes the satellite relative to its three axes:
 - **Roll:** Rotation around the longitudinal axis (tangent to orbit velocity).
 - **Pitch:** Rotation around the lateral axis (perpendicular to orbit plane).
 - **Yaw:** Rotation around the vertical axis (pointing toward Earth's center).
- **Orbit Control (Station Keeping):** Fires thrusters to correct long-term orbital drift caused by gravitational pulls from the Sun/Moon, Earth's equatorial asymmetry (ellipticity), and solar radiation pressure.

3.1 Stabilization Techniques

There are two main methods to keep a satellite from tumbling:

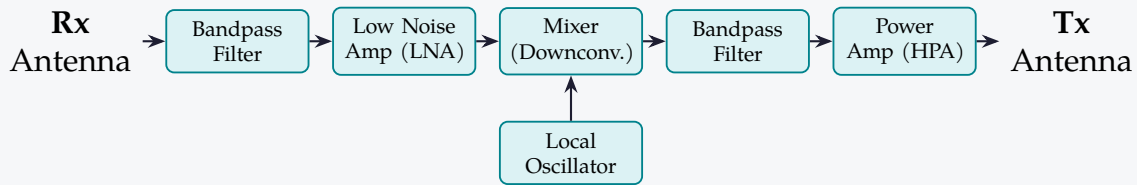
1. **Spin Stabilization:** The satellite body is spun rapidly at 30 to 100 rpm, using gyroscopic momentum to maintain axis alignment. The antennas must be mounted on a "despun" platform rotating in the opposite direction to remain fixed on Earth.
2. **Three-Axis (Body) Stabilization:** The satellite body remains fixed relative to Earth. Momentum wheels or reaction wheels inside the satellite spin at varying speeds to absorb and counteract external disturbance torques.

4. Communication Sub-system: Transponders

The transponder is the heart of the communication payload. A satellite typically houses 12 to 96 transponders, each acting as a separate channel.

4.1 Bent-Pipe (Frequency Translation) Transponder

Also known as a **repeater transponder**. It simply amplifies the incoming signal, down-converts its frequency, and transmits it. It does not look at or alter the digital content.

Bent-Pipe Transponder Block Diagram

- **Low Noise Amplifier (LNA):** Amplifies the extremely weak uplink signal (in the range of picowatts) while adding minimal thermal noise.
- **Mixer & Local Oscillator:** Translates the signal frequency. The shift is $\Delta f = f_{LO} = f_u - f_d$ (e.g., $6\text{ GHz} - 4\text{ GHz} = 2\text{ GHz}$).
- **High Power Amplifier (HPA):** Amplifies the downlink signal to several watts using Traveling-Wave Tube Amplifiers (TWTAs) or Solid-State Power Amplifiers (SSPAs) for transmission.

4.2 Regenerative (Demodulation/Remodulation) Transponder

A regenerative transponder demodulates the incoming analog signal down to the digital baseband bitstream, cleans up the digital noise (performing error correction), and then remodulates the digital data onto a new downlink carrier. It acts as a “**computer in space**”, isolating the uplink noise from the downlink path.

5. Power Sub-system

The power subsystem must generate, store, regulate, and distribute electrical power.

- **Primary Source (Solar Arrays):** Gallium Arsenide (GaAs) or Silicon solar panels convert solar energy into electrical power. In GEO, arrays must continuously track the Sun.
- **Secondary Source (Batteries):** Rechargeable batteries (Nickel-Hydrogen Ni-H₂ or Lithium-Ion) store electrical energy to power the spacecraft during **solar eclipses** when Earth blocks sunlight.
- **Power Conditioning Unit (PCU):** Regulates the bus voltage (typically 28V, 50V, or 100V DC) and controls battery charging/discharge rates.

Quick Revision — Module III

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Key Formulas at a Glance

LO Frequency Shift	$f_{LO} = f_{uplink} - f_{downlink}$
Solar Panel Efficiency	$P = \eta \cdot S \cdot A \cdot \cos(\theta)$ ($S \approx 1367 \text{ W/m}^2$, η = efficiency)
Battery Capacity	$C_{req} = \frac{P_{load} \times t_{eclipse}}{DoD \times V_{bus}}$ (DoD = Depth of Discharge)
Local Oscillator Shift (C-Band)	$f_{LO} = 5.925 \text{ GHz} - 3.7 \text{ GHz} = 2.225 \text{ GHz}$

One-Line Definitions

- **TTC&M:** Ground station radio link monitoring health, status, and position of the satellite.
- **AOCS:** The subsystem stabilizing satellite orientation and correcting orbital drift.
- **Bent-Pipe Transponder:** A simple frequency translation and amplification transponder.
- **Regenerative Transponder:** A transponder that demodulates, processes, and remodulates data.
- **Despun Platform:** A platform rotating oppositely to a spin-stabilized satellite body to keep antennas fixed.
- **Reaction Wheels:** Internal motor-driven wheels adjusting satellite attitude via conservation of angular momentum.
- **Station Keeping:** Maneuvers using thrusters to keep a satellite within its assigned orbital box.

Frequently Asked / Viva Questions

Q1. Differentiate between satellite payload and satellite bus.

The payload represents the hardware performing the actual service (transponders and antennas). The bus represents the supporting infrastructure (structure, power, thermal, AOCS, TTC&M).

Q2. What are the functions of Telemetry and Command in TTC&M?

Telemetry transmits internal sensor readings (health data) from the satellite to Earth. Command receives control signals from Earth to execute adjustments inside the satellite.

Q3. Explain the three attitude axes of a satellite: Roll, Pitch, and Yaw.

Roll: rotation about the velocity vector. Pitch: rotation perpendicular to the orbit plane. Yaw: rotation pointing directly toward the center of the Earth.

Frequently Asked / Viva Questions (contd.)**Q4. Compare Spin Stabilization and Three-Axis Stabilization.**

Spin stabilization uses gyroscopic spin of the entire body to maintain axis alignment. Three-axis stabilization keeps the outer body stationary while internal reaction wheels spin dynamically to absorb disturbance torques.

Q5. Draw the basic block diagram of a bent-pipe transponder.

Rx Antenna → BPF → LNA → Mixer (with LO) → BPF → HPA → Tx Antenna.

Q6. Why is an LNA placed as the first block in the transponder receiver chain?

Because the received uplink signal is extremely weak. Amplifying it first with a Low Noise Amplifier establishes a high signal-to-noise ratio, preventing subsequent stages from corrupting the signal.

Q7. What is the advantage of a regenerative transponder over a bent-pipe transponder?

It demodulates and decodes the signal, stripping away uplink noise and distortion. It then regenerates and remodulates the clean digital bits, preventing noise accumulation.

Q8. Why do satellites require battery systems if they have solar panels?

Because satellites periodically pass through the Earth's shadow (solar eclipses) where solar panels cannot generate power. Batteries provide secondary power during these shadow periods.

Q9. What is the function of the thrusters in the propulsion system?

To perform station-keeping (correcting orbital drift caused by Sun/Moon gravity) and attitude adjustments when reaction wheels saturate.

Q10. What is Depth of Discharge (DoD) in satellite batteries?

It is the percentage of battery capacity used during an eclipse. Satellites restrict DoD to 50–80% to maximize the operational lifespan of the batteries.

Q11. Why is thermal control critical in spacecraft design?

In vacuum, heat is only transferred by radiation. The side facing the Sun heats to $> 100^{\circ}\text{C}$ while the shadow side drops to $< -100^{\circ}\text{C}$. The thermal system redistributes heat to keep electronics in operational ranges.

Q12. What is a local oscillator frequency offset, and how is it chosen?

It is chosen to match the frequency separation between uplink and downlink bands (e.g., 2.225 GHz offset for C-band) to prevent electromagnetic feedback.

Comparison of Transponder Architectures

Feature	Bent-Pipe Transponder	Regenerative Transponder
Basic Operation	RF-to-RF translation	Demodulation → Baseband Processing → Remodulation
Noise Propagation	Uplink noise is amplified and sent down.	Uplink noise is removed at baseband.
Complexity & Weight	Low complexity, lightweight, highly reliable.	High complexity, heavy, requires onboard processors.
Flexibility	Protocol transparent (handles any modulation format).	Protocol-specific (locked to a particular modulation/coding standard).
Primary Use Case	Standard satellite TV distribution, VSAT links.	Modern satellite broadband, military secure links.